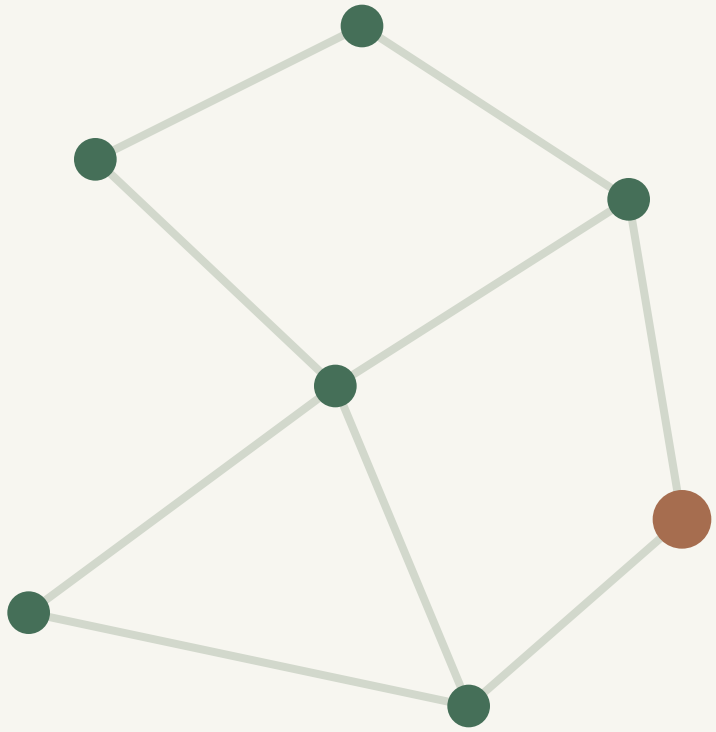
RESEARCH BRIEFING | OCTOBER 2026

Open-World Network-Flow Triage



Can a fixed review budget surface attack categories that were never represented in training?

A TRANSPARENT BENCHMARK ON UNSW-NB15

THE QUESTION

The benchmark isolates a real tension

Known-risk ranking can be precise while missing behavior outside its training labels.

73,791
primary test flows

10%
primary review budget

9
attack-category holdouts

MAIN READOUT

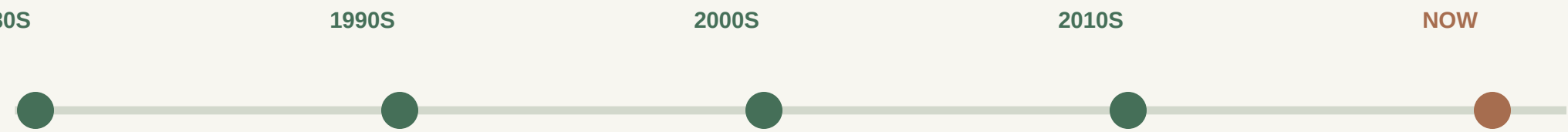
Novelty-only ranking found the most held-out attack examples, but sent more normal flows for review. The 75/25 two-lane queue reduced that workload and recovered known-category performance, while finding fewer held-out examples.

This is an evaluation of ranking policies on a cyber-range dataset, not proof of zero-day detection or a new detection algorithm.

PATTERN HISTORY

Network detection added context over time

A teaching timeline: each approach answers a different question, and all remain in use.



Behavior models

Does activity depart from a user or system baseline?

Signatures and rules

Does an event match a known indicator or condition?

Statistical learning

Can distributions and learned boundaries separate normal from suspicious?

Supervised ML

Can labeled examples rank likely attacks?

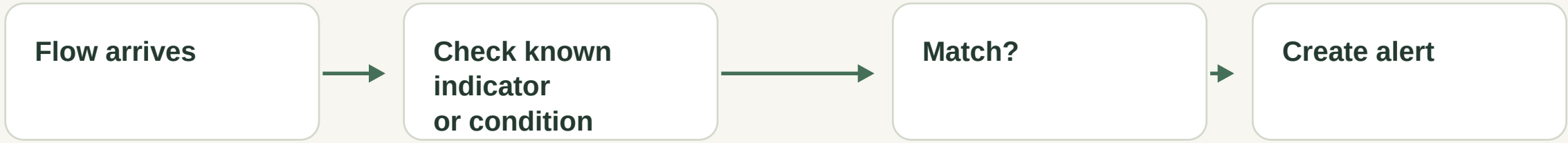
Open-world and adaptive IDS

What happens when behavior or labels change?

The timeline summarizes recurring research and engineering patterns; it does not imply that one replaced another.

Pattern 1 | Signatures and explicit rules

DECISION PATH



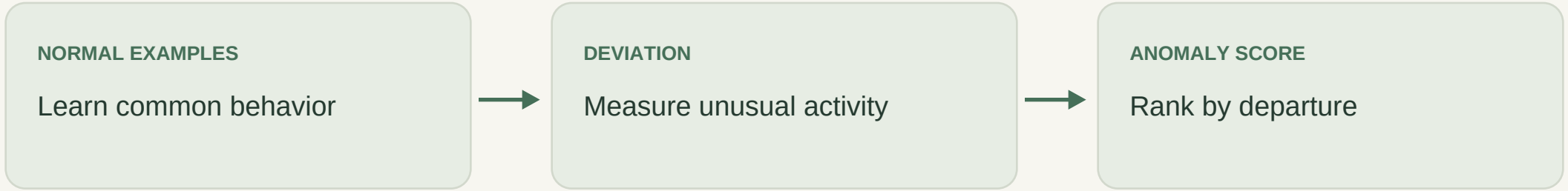
- Readable and fast to investigate when the behavior is specified.
- Coverage depends on known patterns and conditions being current.

Examples in practice include signature rules in network IDS such as Snort and configurable event rules in security analytics platforms.

REFERENCE Roesch (1999), Snort - Lightweight Intrusion Detection for Networks, USENIX LISA.

Pattern 2 | Anomaly detection

BASELINE LEARNING

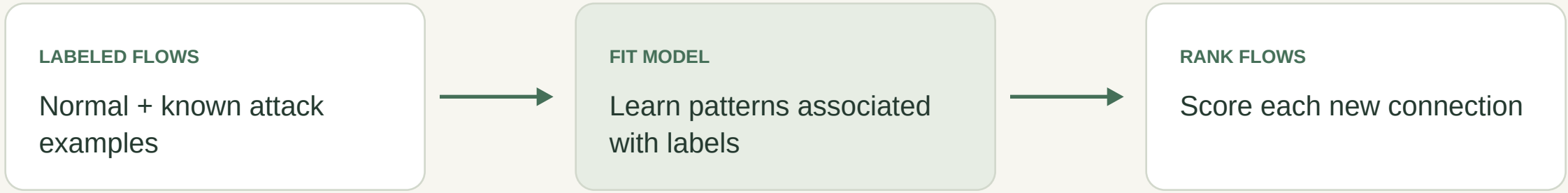


An anomaly is evidence of difference, not proof of malicious intent. The question depends on what data formed the baseline.

In this benchmark, an Isolation Forest is fit only on normal-labeled training flows and produces a novelty ranking.

REFERENCE Denning (1987), An Intrusion-Detection Model, IEEE Transactions on Software Engineering.

Pattern 3 | Supervised attack scoring

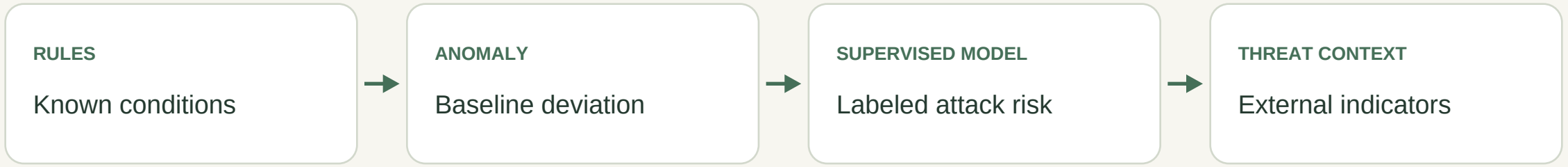


Supervised models can rank represented patterns effectively. Their known classes and labels define what “attack” means during training.

The open-world question begins where the label set stops: how does a queue surface behavior outside those examples?

Pattern 4 | Layered detection systems

COMMON SIGNAL SOURCES



Products often combine multiple detection mechanisms. The useful design question is not whether one signal is universally best, but how evidence is prioritized and reviewed.

The project treats the two score types as separate lanes to make their workload and coverage trade-offs visible.

Current research | Open-world recognition and adaptation

The field is moving beyond a fixed set of known labels.

01 **Unknown recognition**

Identify examples that do not fit known classes.

02 **Inspection under budget**

Use limited expert review to characterize unfamiliar behavior.

03 **Continual learning**

Update models as labels and traffic patterns change.

04 **Distribution shift**

Measure when the data no longer resemble development traffic.

These are active research areas, not a claim that the project invents them. This MVP focuses on one transparent, fixed-budget comparison.

RESEARCH EXAMPLES Open Recognition and Inspection (IEEE, 2025) | SPIDER continual-learning NIDS (IEEE INFOCOM, 2024)

STUDY DESIGN

The project question

How should a limited review queue behave when a category was excluded from training?

Closed-set question

“Which known attacks look risky?”

Open-world question

“Does the queue surface a category the model never saw?”

Benchmark proxy: one labeled UNSW-NB15 attack category is withheld per fold.

“Unseen” means withheld category in this dataset. It does not mean a true zero-day event.

BENCHMARK POPULATION

Data | UNSW-NB15

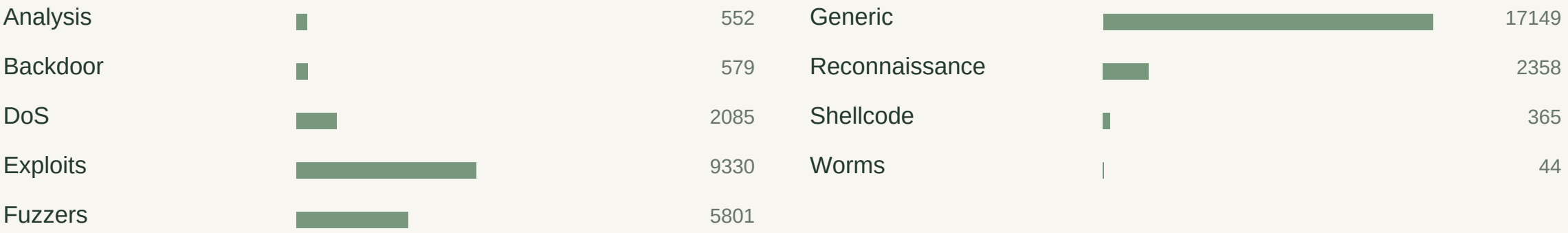
A cyber-range collection of normal and simulated attack network flows.

175,341
published training flows

82,332
published test flows

73,791
primary test after exact-match removal

ATTACK CATEGORIES



Counts are attack-category support in the primary test set. The dataset is attack-heavy and does not represent a production network base rate.

Data | Features and target boundaries

Twenty model input columns; target and category fields stay outside the feature matrix.

12 TRAFFIC MEASUREMENTS

dur, spkts, dpkts, sbytes, dbytes, rate, sttl, dttl, sload, dload, sloss, dloss

5 RECENT-CONNECTION COUNTS

ct_srv_src, ct_dst_ltm, ct_src_dport_ltm, ct_dst_sport_ltm, ct_dst_src_ltm

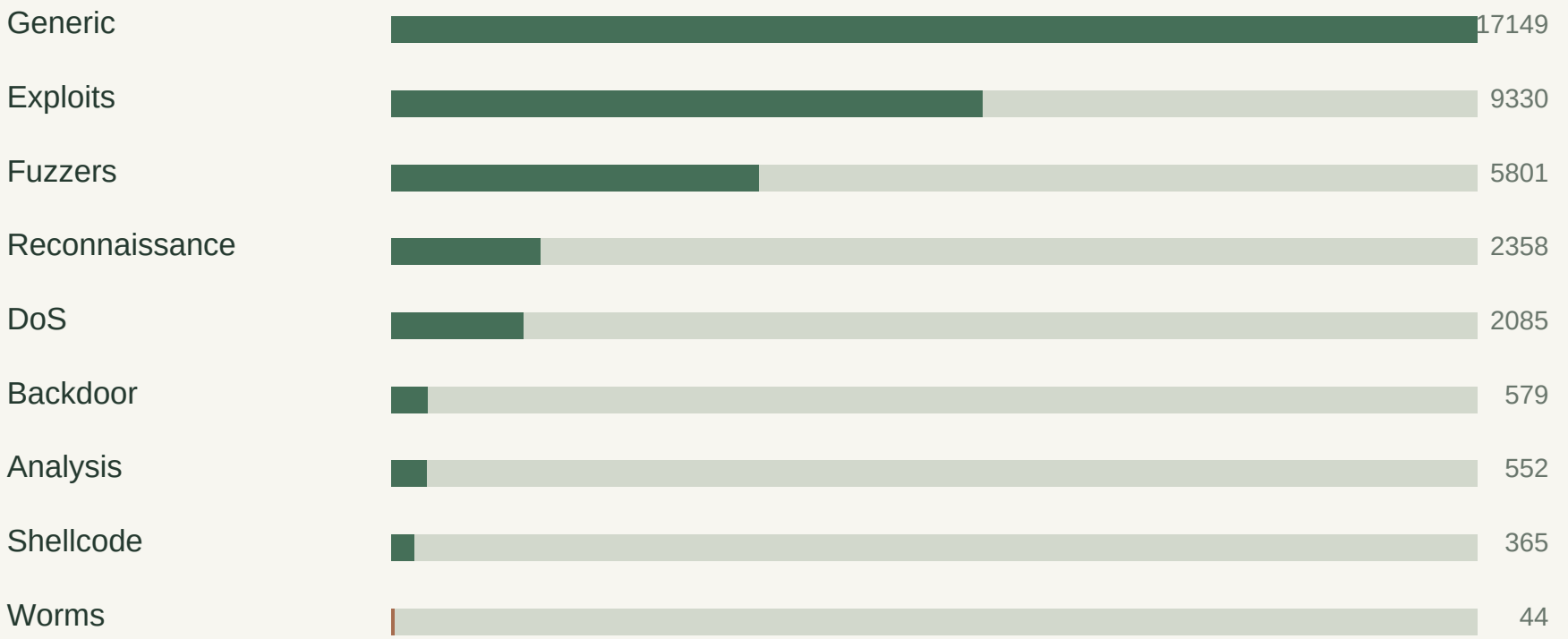
3 CONNECTION CATEGORIES

proto, service, state

EXCLUDED FROM MODEL INPUTS label = binary outcome | attack_cat = fold and report only | id = row identifier

The nine held-out categories

Each fold removes one category from training, then checks its ranking in the same test population.



SMALLEST CLASS

44

Worms flows

Reported separately; excluded from macro averages.

Low support makes category recall noisy, especially for Worms and Shellcode. Category-level results accompany the macro average.

Study design | Six reproducible steps

The held-out test labels are revealed only after the rankings are built.

01 **Audit exact overlap**

Remove exact train/test feature matches from primary test.

02 **Choose a held-out category**

Repeat once for each of nine attack categories.

03 **Fit known-risk model**

Train XGBoost on normal + remaining attack categories.

04 **Fit novelty model**

Train Isolation Forest using normal flows only.

05 **Rank within the budget**

Build the three equal-size review queues.

06 **Score after selection**

Measure recall, precision, and benign review volume.

System design | Separate scores, shared budget

The two scores answer different questions and are not treated as calibrated probabilities.

KNOWN-RISK SCORE

XGBoost trained on normal and seen attacks

NOVELTY SCORE

Isolation Forest fitted only on normal flows

FIXED-SIZE QUEUE

At each budget, select exactly the same number of test flows for every policy.

5% | 10% primary | 20%

Three policies at the same review volume

POLICY 01

Known-risk only

100% known-risk ranking

Best fit to seen attack labels

POLICY 02

Novelty only

100% novelty ranking

Explore behavior unlike normal flows

POLICY 03

Two-lane 75/25

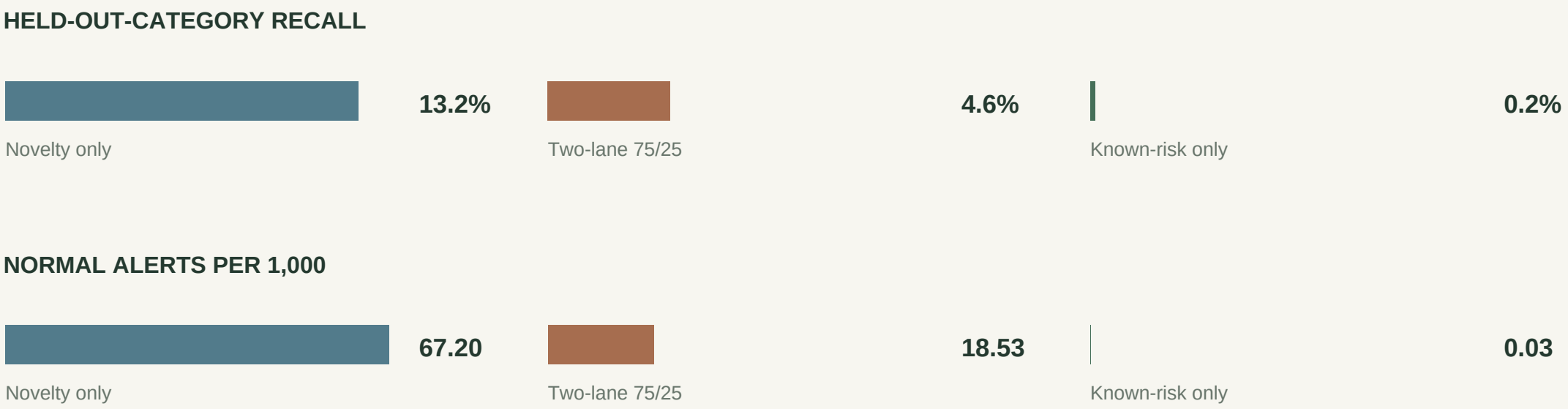
75% known risk + 25% novelty

Reserve a lane for unfamiliar behavior

The 75/25 split is an illustrative allocation. It is not tuned or presented as an industry standard.

Primary result | A trade-off, not a winner

Macro average across eight categories with at least 100 primary-test examples; 10% review budget.



Novelty-only leads held-out recall, with more benign review. The two-lane policy trades some unknown-category coverage for fewer normal alerts and stronger known-category recall.

Primary budget | Complete policy comparison

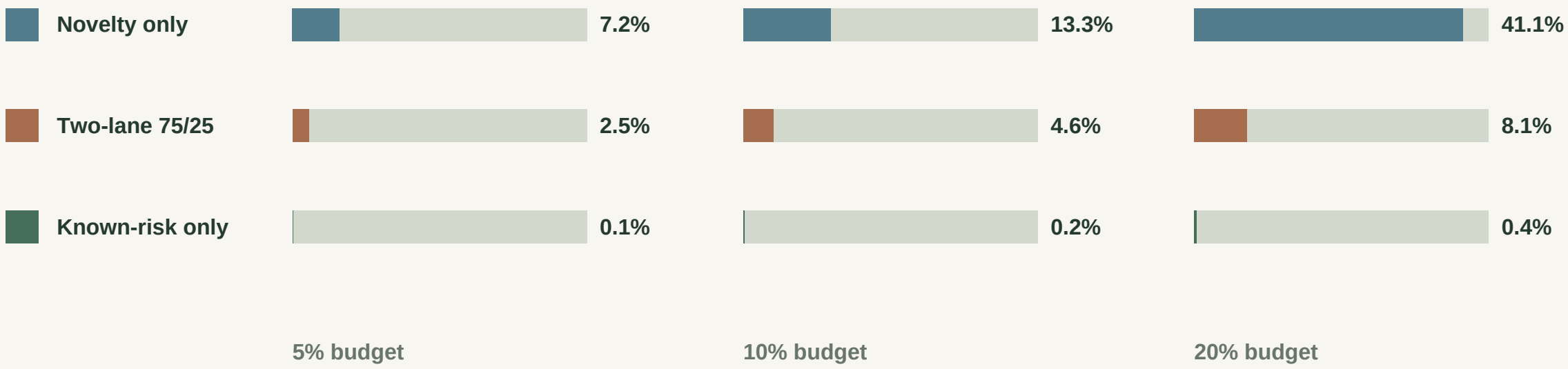
10% review budget; eight-category macro averages.

Policy	Unknown recall	Known recall	Precision	Normal / 1,000
Known-risk only	0.2%	22.7%	100.0%	0.03
Novelty only	13.3%	13.1%	67.6%	67.2
Two-lane 75/25	**4.6%**	**20.1%**	**91.1%**	**18.5**

Precision and benign-alert rates are specific to this attack-heavy benchmark. Do not interpret them as production SOC forecasts.

Review budget sensitivity

Supported-category macro recall; eight categories, equal category weights.



At larger budgets, novelty ranking surfaces more withheld-category examples; benign review also rises. Two-lane routing remains between novelty-only and known-risk-only.

INTERPRETATION

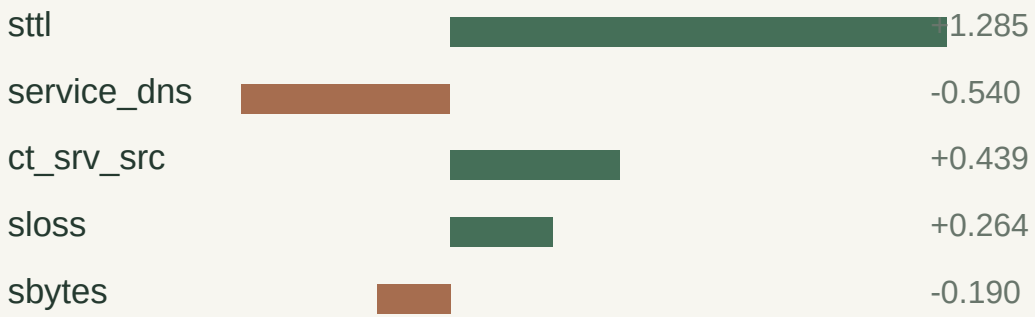
What one flow explanation can and cannot say

The example is a held-out Exploits flow ranked by the known-risk model.

0.8452
XGBoost known-risk score

0.4596
Isolation Forest novelty score

LARGEST LOCAL XGBOOST CONTRIBUTIONS



SHAP contributions explain the XGBoost score for this one flow. The percentile comparison gives context for unusual values; it is not an explanation of Isolation Forest or attacker intent.

Interpretation and next research questions

- Withheld categories are a benchmark proxy for unknown behavior, not zero-day attacks.
- No timestamps means no temporal drift or future-period evaluation.
- UNSW-NB15 is cyber-range generated and attack-heavy; production precision will differ.
- Worms has 44 primary-test flows; exact duplicate filtering cannot remove all dependence.
- The 75/25 lane split is a baseline policy; test alternative allocations and calibration on a distinct development set.

REFERENCES

Moustafa & Slay (2015), UNSW-NB15, DOI 10.1109/MilCIS.2015.7348942.

Toward Open-World Network Intrusion Detection via Open Recognition and Inspection, IEEE 2025: ieeexplore.ieee.org/document/11159092/

SPIDER continual-learning NIDS, IEEE INFOCOM 2024: newslab.iith.ac.in/publications/spider-a-semi-supervised-continual-learning-based-network-intrusion-detection-system/